AI-Assisted Adverse Drug Reaction (ADR) Detection Tool

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Problem Statement

Pharmacovigilance systems rely on accurate and timely identification of adverse drug reactions (ADRs). Manual analysis of case reports can be time-consuming and prone to variability. This project explores how generative AI can assist in identifying potential ADRs and categorizing their severity.

Objective

To develop a simple AI-assisted prototype that analyzes drug-symptom inputs and provides potential ADRs, severity classification, and recommended action.

Methodology

1. Used generative AI (LLM-based prompts) to simulate pharmacovigilance reasoning.
2. Input: Drug name and patient symptoms.
3. Output: Possible ADR, severity (mild/moderate/severe), and suggested action.
4. Applied clinical reasoning principles based on pharmacovigilance practices (MedDRA, WHO-UMC concepts).

Sample Case Analysis

Drug	Symptoms	AI Output (ADR & Severity)	Recommended Action
Paracetamol	Skin rash	Mild hypersensitivity	Monitor
Amoxicillin	Swelling, itching	Allergic reaction (Moderate)	Stop drug, consult doctor
Clozapine	Fever, fatigue	Possible agranulocytosis (Severe)	Immediate evaluation
Ibuprofen	Gastric pain	GI irritation (Mild)	Take with food
Metformin	Nausea, diarrhea	GI intolerance (Mild)	Dose adjustment

Results & Insights

The AI model was able to generate clinically relevant ADR interpretations and categorize severity with reasonable accuracy for simulated cases. It demonstrated potential to assist in early signal detection and decision support.

Limitations

This is a prototype based on simulated inputs. Outputs depend on prompt design and may not replace clinical judgment. Real-world validation is required.

Conclusion

AI-assisted tools can enhance pharmacovigilance workflows by supporting ADR detection and clinical decision-making. Further development can integrate real-world data for scalable applications.